A FINITE TYPE ALGEBRA OF RELATIVE DIMENSION N IS QUASI-FINITE OVER A POLYNOMIAL ALGEBRA IN N VARIABLES AFTER INVERTING ONE ELEMENT

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The fibre ring as a finite type algebra over the residue field. Used at the start of the counted proof (lines 31397-31400). Turns the relative dimension hypothesis into a statement about the fibre, where Noether normalisation applies.
- Quasi-finiteness is an open condition. Invoked at the end of the counted proof (lines 31422-31423). Spreads the quasi-finiteness obtained at one point out to a basic open.

Source reference. Original lemma 00OY: sources/stacks-algebra/pinned-original-source.tex, source lines 28242–28242. Navigation pointer for \ref{lemma-Noether-normalization} used inside the counted span (line 31416); the argument that consumes it is retained in the span.

Date: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00QA: sources/stacks-algebra/pinned-original-source.tex, source lines 31097–31097. Navigation pointer for \ref{lemma-quasi-finite-open} used inside the counted span (line 31422); the argument that consumes it is retained in the span.

Source reference. Original lemma 00E6: sources/stacks-algebra/pinned-original-source.tex, source lines 3639–3639. Navigation pointer for \ref{remark-fundamental-diagram} used inside the counted span (line 31407); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $R \to S$ be a finite type ring map. Let $\mathfrak{q} \subset S$ be a prime. Suppose that $\dim_{\mathfrak{q}}(S/R) = n$. There exists a $g \in S$, $g \notin \mathfrak{q}$ such that S_g is quasi-finite over a polynomial algebra $R[t_1, \ldots, t_n]$.*

Proof. The ring $\overline{S} = S \otimes_R \kappa(\mathfrak{p})$ is of finite type over $\kappa(\mathfrak{p})$. Let $\overline{\mathfrak{q}}$ be the prime of $\overline{S}$ corresponding to $\mathfrak{q}$. By definition of the dimension of a topological space at a point there exists an open $U \subset \mathrm{Spec}(\overline{S})$ with $\overline{q} \in U$ and $\dim(U) = n$. Since the topology on $\mathrm{Spec}(\overline{S})$ is induced from the topology on $\mathrm{Spec}(S)$ (see Remark 00E6), we can find a $g \in S$, $g \notin \mathfrak{q}$ with image $\overline{g} \in \overline{S}$ such that $D(\overline{g}) \subset U$. Thus after replacing S by S_g we see that $\dim(\overline{S}) = n$.

Next, choose generators $x_1, \ldots, x_N$ for S as an R-algebra. By Lemma 00OY there exist elements $y_1, \ldots, y_n$ in the $\mathbf{Z}$-subalgebra of S generated by $x_1, \ldots, x_N$ such that the map $R[t_1, \ldots, t_n] \to S$, $t_i \mapsto y_i$ has the property that $\kappa(\mathfrak{p})[t_1 \ldots, t_n] \to \overline{S}$ is finite. In particular, S is quasi-finite over $R[t_1, \ldots, t_n]$ at $\mathfrak{q}$. Hence, by Lemma 00QA we may replace S by S_g for some $g \in S$, $g \notin \mathfrak{q}$ such that $R[t_1, \ldots, t_n] \to S$ is quasi-finite. $\square$